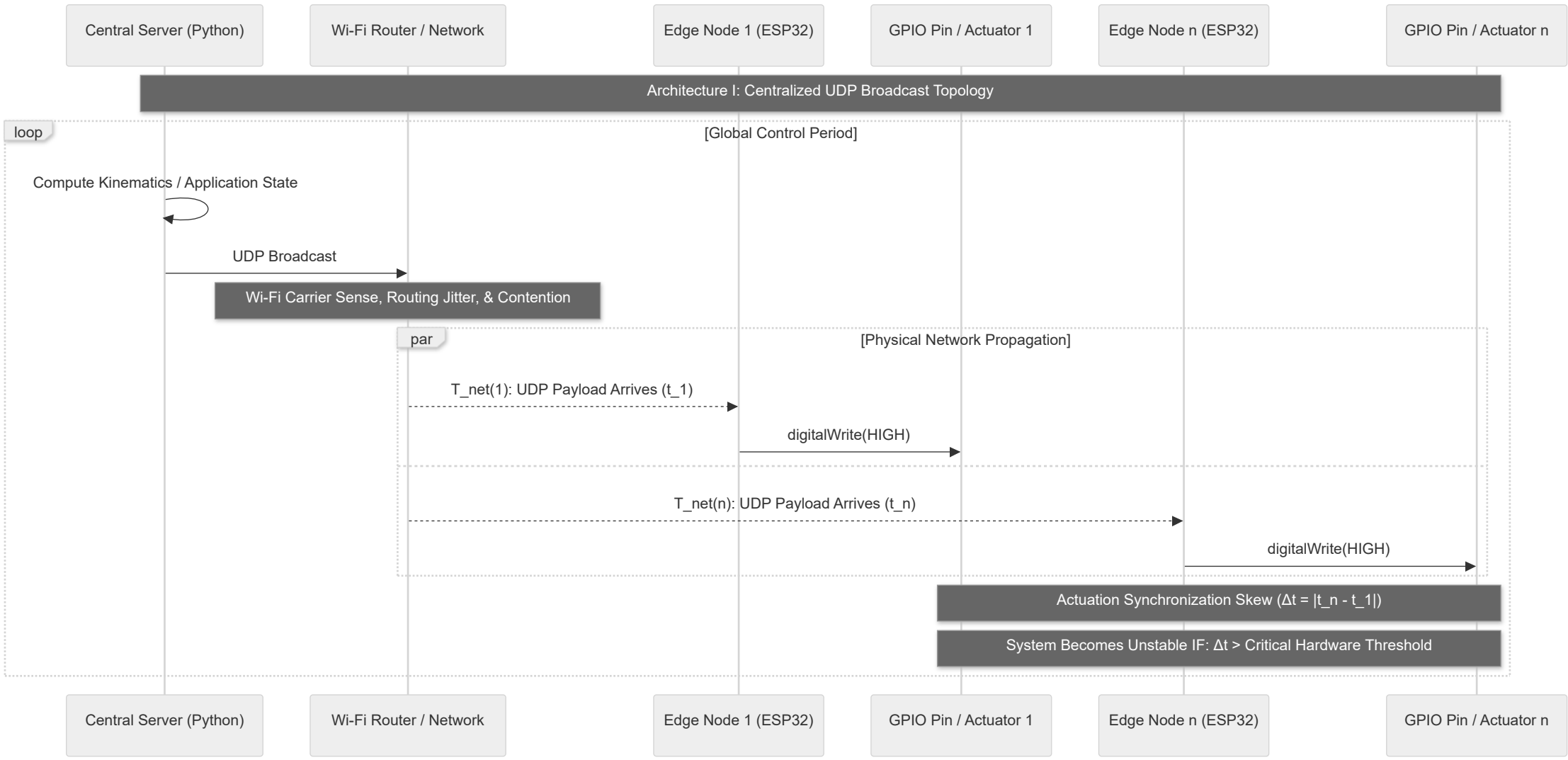
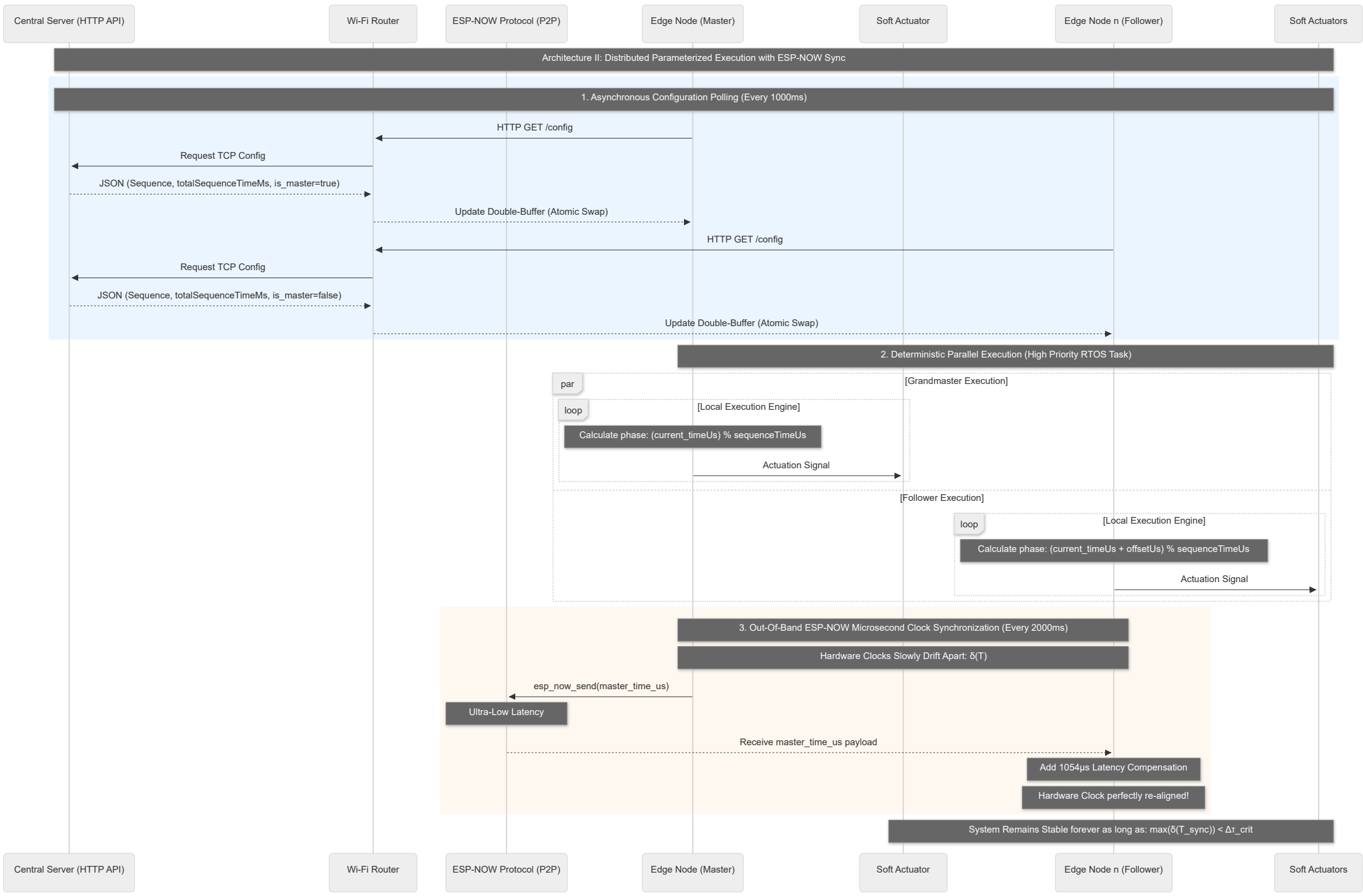
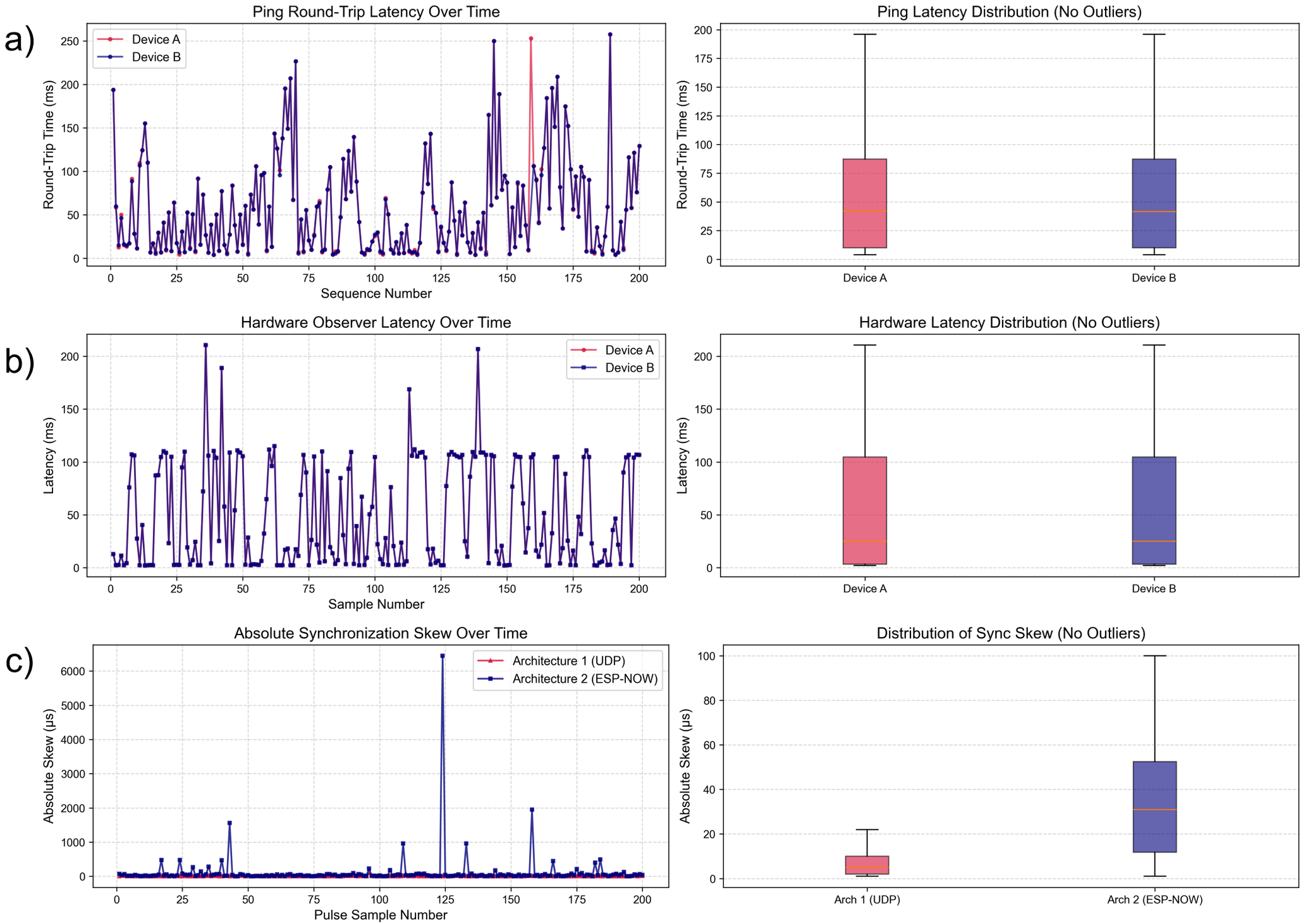
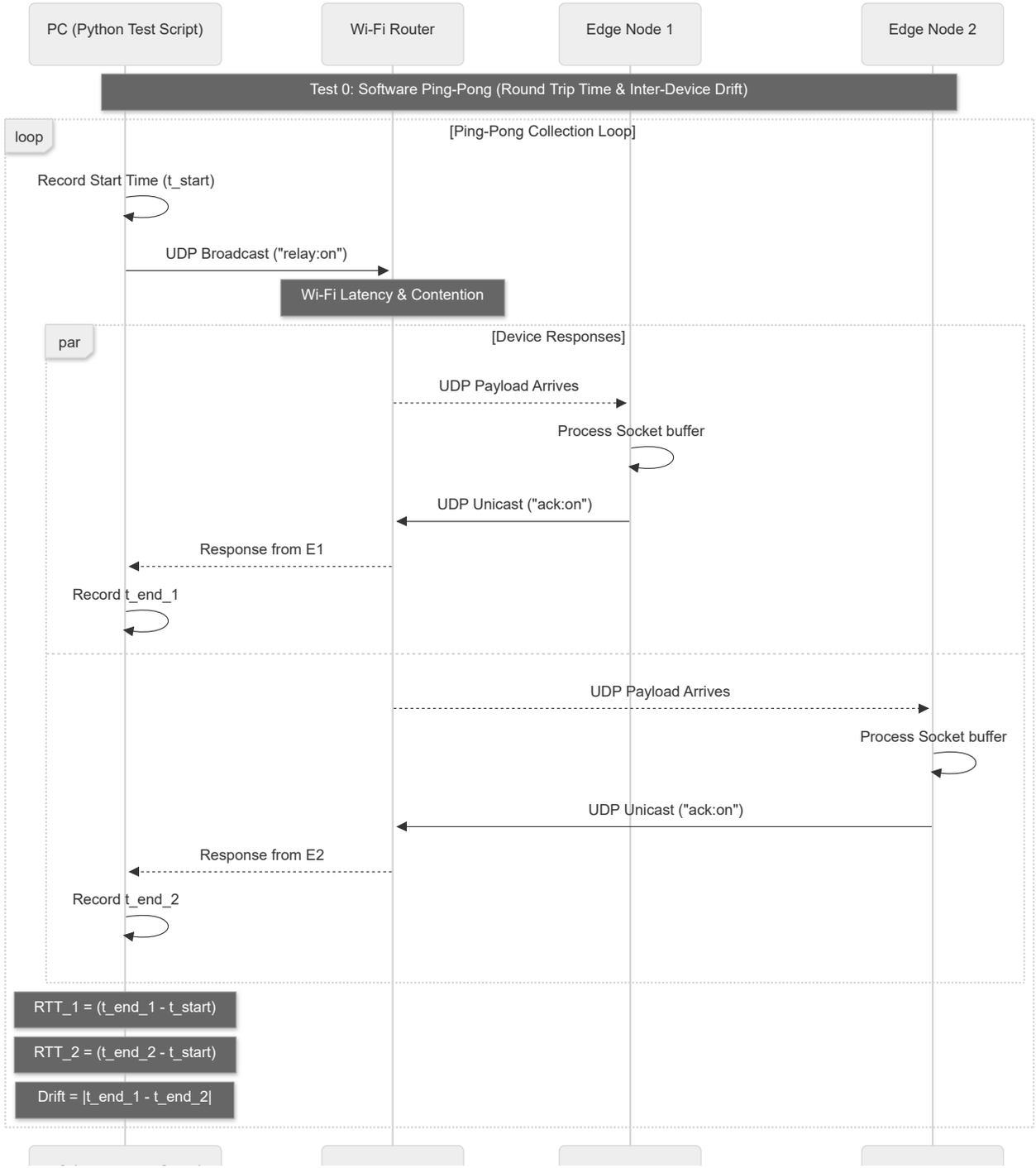
a)

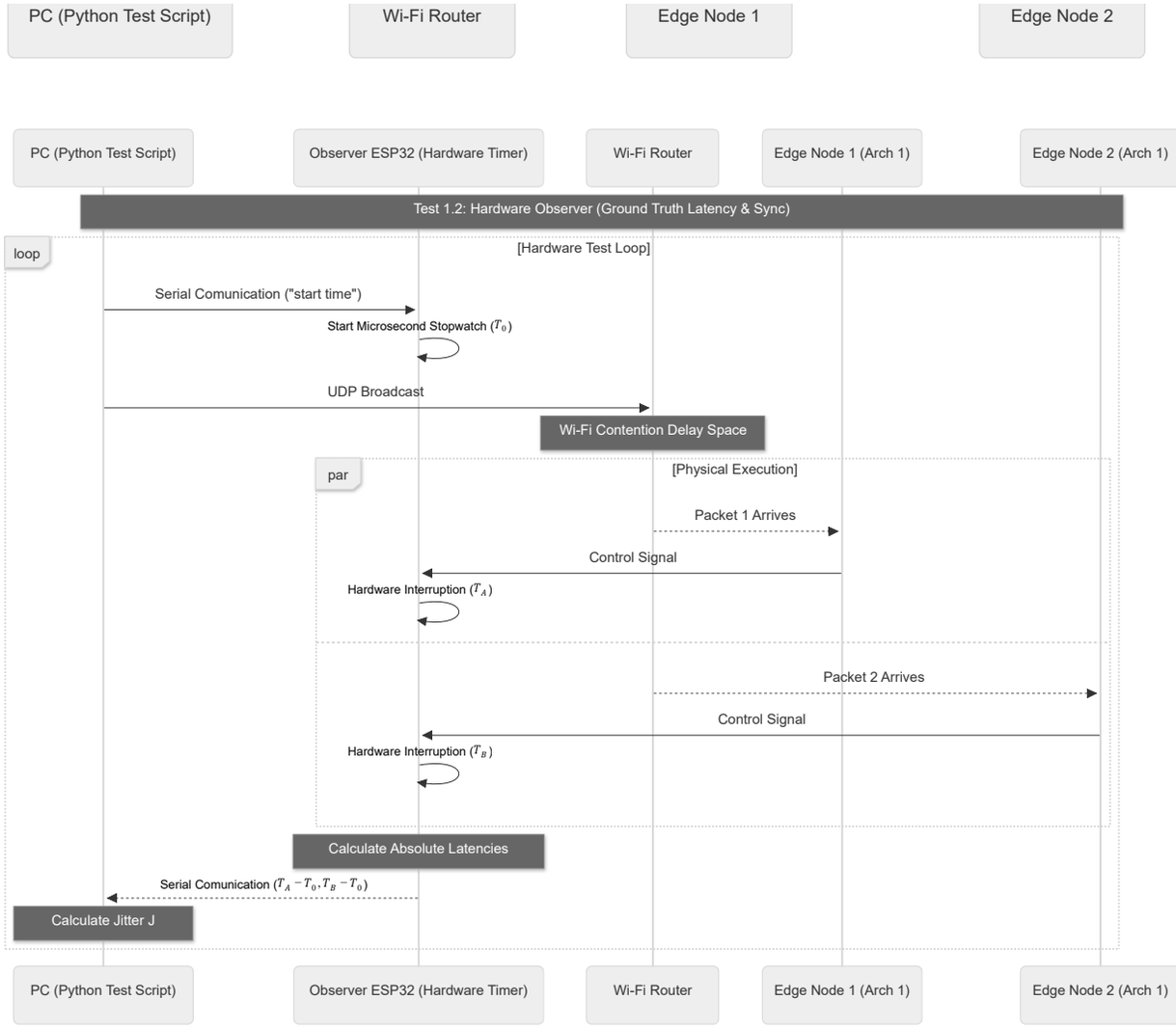

Central Server (Python)
Wi-Fi Router / Network
Edge Node 1 (ESP32)
GPIO Pin / Actuator 1
Edge Node n (ESP32)
GPIO Pin / Actuator n
Architecture I: Centralized UDP Broadcast Topology
loop
[Global Control Period]
Compute Kinematics / Application State
UDP Broadcast
Wi-Fi Carrier Sense, Routing Jitter, & Contention
par
[Physical Network Propagation]
T_net(1): UDP Payload Arrives (t_1)
digitalWrite(HIGH)
T_net(n): UDP Payload Arrives (t_n)
digitalWrite(HIGH)
Actuation Synchronization Skew (Δt = |t_n - t_1|)
System Becomes Unstable IF: Δt > Critical Hardware Threshold
Central Server (Python)
Wi-Fi Router / Network
Edge Node 1 (ESP32)
GPIO Pin / Actuator 1
Edge Node n (ESP32)
GPIO Pin / Actuator n

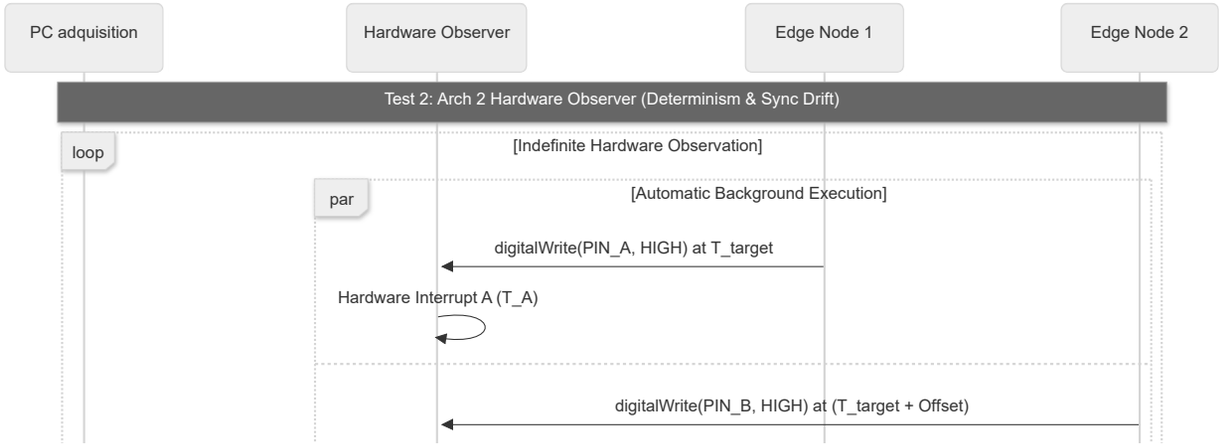
b)

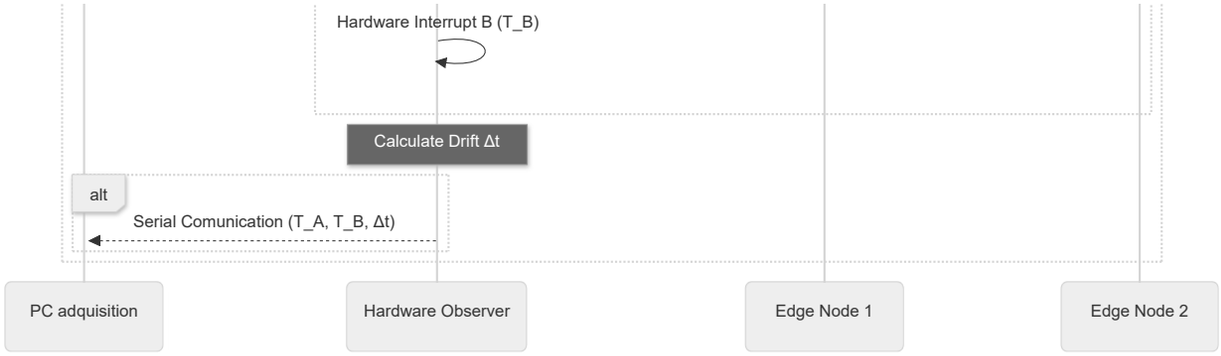

Central Server (HTTP API)
Wi-Fi Router
ESP-NOW Protocol (P2P)
Edge Node (Master)
Soft Actuator
Edge Node n (Follower)
Soft Actuators
Architecture II: Distributed Parameterized Execution with ESP-NOW Sync
1. Asynchronous Configuration Polling (Every 1000ms)
HTTP GET /config
Request TCP Config
JSON (Sequence, totalSequenceTimeMs, is_master=true)
Update Double-Buffer (Atomic Swap)
HTTP GET /config
Request TCP Config
JSON (Sequence, totalSequenceTimeMs, is_master=false)
Update Double-Buffer (Atomic Swap)
2. Deterministic Parallel Execution (High Priority RTOS Task)
par
[Grandmaster Execution]
loop
[Local Execution Engine]
Calculate phase: (current_timeUs) % sequenceTimeUs
Actuation Signal
[Follower Execution]
loop
[Local Execution Engine]
Calculate phase: (current_timeUs + offsetUs) % sequenceTimeUs
Actuation Signal
3. Out-Of-Band ESP-NOW Microsecond Clock Synchronization (Every 2000ms)
Hardware Clocks Slowly Drift Apart: δ(T)
esp_now_send(master_time_us)
Ultra-Low Latency
Receive master_time_us payload
Add 1054µs Latency Compensation
Hardware Clock perfectly re-aligned!
System Remains Stable forever as long as: max(δ(T_sync)) < Δτ_crit
Central Server (HTTP API)
Wi-Fi Router
ESP-NOW Protocol (P2P)
Edge Node (Master)
Soft Actuator
Edge Node n (Follower)
Soft Actuators


a)
Ping Round-Trip Latency Over Time
Device A
Device B
Round-Trip Time (ms)
Sequence Number
Ping Latency Distribution (No Outliers)
Round-Trip Time (ms)
Device A
Device B
b)
Hardware Observer Latency Over Time
Device A
Device B
Latency (ms)
Sample Number
Hardware Latency Distribution (No Outliers)
Latency (ms)
Device A
Device B
c)
Absolute Synchronization Skew Over Time
Architecture 1 (UDP)
Architecture 2 (ESP-NOW)
Absolute Skew (µs)
Pulse Sample Number
Distribution of Sync Skew (No Outliers)
Absolute Skew (µs)
Arch 1 (UDP)
Arch 2 (ESP-NOW)


PC (Python Test Script)
Wi-Fi Router
Edge Node 1
Edge Node 2
Test 0: Software Ping-Pong (Round Trip Time & Inter-Device Drift)
loop
[Ping-Pong Collection Loop]
Record Start Time (t_start)
UDP Broadcast ("relay:on")
Wi-Fi Latency & Contention
par
[Device Responses]
UDP Payload Arrives
Process Socket buffer
UDP Unicast ("ack:on")
Response from E1
Record t_end_1
UDP Payload Arrives
Process Socket buffer
UDP Unicast ("ack:on")
Response from E2
Record t_end_2
RTT_1 = (t_end_1 - t_start)
RTT_2 = (t_end_2 - t_start)
Drift = |t_end_1 - t_end_2|


PC (Python Test Script)
Wi-Fi Router
Edge Node 1
Edge Node 2
PC (Python Test Script)
Observer ESP32 (Hardware Timer)
Wi-Fi Router
Edge Node 1 (Arch 1)
Edge Node 2 (Arch 1)
Test 1.2: Hardware Observer (Ground Truth Latency & Sync)
loop
[Hardware Test Loop]
Serial Comunication ("start time")
Start Microsecond Stopwatch (T_0)
UDP Broadcast
Wi-Fi Contention Delay Space
par
[Physical Execution]
Packet 1 Arrives
Control Signal
Hardware Interruption (T_A)
Packet 2 Arrives
Control Signal
Hardware Interruption (T_B)
Calculate Absolute Latencies
Serial Comunication ($T_A - T_0, T_B - T_0$)
Calculate Jitter J
PC (Python Test Script)
Observer ESP32 (Hardware Timer)
Wi-Fi Router
Edge Node 1 (Arch 1)
Edge Node 2 (Arch 1)


PC adquisition
Hardware Observer
Edge Node 1
Edge Node 2
Test 2: Arch 2 Hardware Observer (Determinism & Sync Drift)
loop
[Indefinite Hardware Observation]
par
[Automatic Background Execution]
digitalWrite(PIN_A, HIGH) at T_target
Hardware Interrupt A (T_A)
digitalWrite(PIN_B, HIGH) at (T_target + Offset)


Hardware Interrupt B (T_B)
Calculate Drift Δt
alt
Serial Comunication (T_A, T_B, Δt)
PC adquisition
Hardware Observer
Edge Node 1
Edge Node 2